

CropSage Project Phases

Current scope, completed foundation work, accepted limitations and the path to an end-to-end crop recommendation.

PHASE 01 Foundation COMPLETE	PHASE 02 Evidence normalization NEXT	PHASE 03 Decision engine IN PROGRESS	PHASE 04 Application integration PLANNED	PHASE 05 End-to-end validation PLANNED
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Phase 1 - Farm Profile and Crop Catalog
Requirements and evidence foundations for preliminary crop suitability in Plainview, Texas.

COMPLETE

Farm Profile Foundation

- 1. Required location and planting-time inputs are defined.
- 2. Farmer irrigation availability and reliability are documented, with optional soil overrides.
- 3. Missing optional evidence reduces confidence instead of becoming a favorable match.
- 4. The demonstration farm is fixed at Plainview (34.18, -101.76) in the Texas Plains context.

Crop Catalog Foundation

- 1. Catalog version 1.1.0 contains 22 Texas crops and 29 required properties.
- 2. Temperature, soil, water, region, season and provenance fields are documented.
- 3. Nullable evidence policy prevents absent values from being treated as zero or favorable.
- 4. Every crop record remains provisional pending external agronomic review.

Location Evidence Prepared

Provider	Prepared evidence role
NASA POWER	Climate history and climatology for temperature and rainfall context.
FortyGuard	Local heat, threshold exceedance and persistence evidence.
Open-Meteo	Current, recent and forecast weather plus modeled soil moisture.
USDA SSURGO	Plainview soil properties, root-zone limits and available water storage; unified evidence exported.

Phase 1 outcome

The Plainview MVP now has a defined farm profile, a complete crop-requirement catalog and mapped evidence responsibilities. Production normalization and scoring integration remain next.

Accepted MVP Limitations

These constraints are intentional project decisions. They should remain visible in implementation, testing and user-facing interpretation.

1	Plainview is manually assigned to the Texas Plains; no statewide region resolver.
2	Frost-free-season comparison is unavailable and must remain unknown or excluded.
3	Heat thresholds, duration and numeric water ranges remain supplementary under the new policy.
4	Records remain provisional pending external agronomic review.
5	Exact farm boundaries, geocoding, well/canal capacity and farmer soil-depth overrides are optional.
6	Recommendations apply to preliminary suitability, not yield or irrigation prescriptions.

Remaining Implementation Work

PHASE 2	Fix and generate normalized NASA and Open-Meteo evidence exports. Preserve or regenerate sanitized FortyGuard evidence. Combine provider outputs into an evidence bundle.
Evidence normalization	
PHASE 3	Finish the deterministic engine.
Decision engine	
PHASE 4	Build production adapters, application flow and UI.
Product integration	
PHASE 5	Run the final end-to-end Plainview recommendation.
End-to-end validation	

NEXT MILESTONE	A validated, normalized evidence bundle ready for deterministic scoring.
MVP FINISH LINE	A reproducible Plainview run that produces preliminary crop suitability recommendations with reasons, risks and confidence.

Document basis: approved CropSage MVP decisions and current Plainview implementation status.